

$MaxVersion(versions, max_cts) \triangleq$

LET $valid \triangleq \{v \in versions : v.cts \leq max_cts\}$

IN CHOOSE $v \in valid : \forall v2 \in valid : v.cts \geq v2.cts$

$ExecuteOp(t) \triangleq$

$\wedge tx_status[t] = \text{"Active"}$

$\wedge pc[t] \leq Len(Workload[t].ops)$

LET $op \triangleq Workload[t].ops[pc[t]]$

IN IF $op.type = \text{"R"}$

THEN LET $is_in_buffer \triangleq tx_buffer[t][op.key] \neq \langle \rangle$

$latest_ver \triangleq MaxVersion(store[op.key], tx_snapshot[t])$

$read_val \triangleq$ IF is_in_buffer

THEN $tx_buffer[t][op.key][1]$

ELSE $latest_ver.val$

IN $\wedge op_seq' = [op_seq \text{ EXCEPT } ![t] =$

$Append(@, [type \mapsto \text{"R"}, key \mapsto op.key, val \mapsto read_val])]$

$\wedge client_dep' =$

IF $\neg is_in_buffer \wedge$

$latest_ver.cts > client_dep[Workload[t].session]$

THEN $[client_dep \text{ EXCEPT }$

$![Workload[t].session] = latest_ver.cts]$

ELSE $client_dep$

$\wedge tx_checkset' =$

IF $Workload[t].level = \text{"SER"}$

THEN $[tx_checkset \text{ EXCEPT } ![t] = tx_checkset[t] \cup \{op.key\}]$

ELSE $tx_checkset$

$\wedge pc' = [pc \text{ EXCEPT } ![t] = pc[t] + 1]$

$\wedge \text{UNCHANGED } \langle client_last_commit, shard_ct, tx_status,$

$tx_snapshot, tx_buffer, tx_commit_time, store \rangle$

ELSE $op.type = \text{"W"}$

$\wedge tx_buffer' = [tx_buffer \text{ EXCEPT } ![t][op.key] = \langle op.val \rangle]$

$\wedge tx_checkset' =$

IF $Workload[t].level \in \{\text{"PSI"}, \text{"SI"}, \text{"SER"}\}$

THEN $[tx_checkset \text{ EXCEPT } ![t] = tx_checkset[t] \cup \{op.key\}]$

ELSE $tx_checkset$

$\wedge op_seq' = [op_seq \text{ EXCEPT } ![t] =$

$Append(@, [type \mapsto \text{"W"}, key \mapsto op.key, val \mapsto op.val])]$

$\wedge pc' = [pc \text{ EXCEPT } ![t] = pc[t] + 1]$

$\wedge \text{UNCHANGED } \langle client_dep, client_last_commit, shard_ct,$

$tx_status, tx_snapshot, tx_commit_time, store \rangle$